

LAB EXPERIMENTS

Q22) Write a C program for Encrypt and decrypt in cipher block chaining mode using one of the following ciphers: affine modulo 256, Hill modulo 256, S-DES, DES. Test data for S-DES using a binary initialization vector of 1010 1010. A binary plaintext of 0000 0001 0010 0011 encrypted with a binary key of 01111 11101 should give a binary plaintext of 1111 0100 0000 1011. Decryption should work correspondingly.

Aim:

To write a C program to **encrypt and decrypt data using Cipher Block Chaining (CBC) mode with the Affine Cipher modulo 256.**

Algorithm:

1. Start the program.
2. Read the plaintext from the user.
3. Enter the multiplicative key **a**, additive key **b**, and Initialization Vector (IV).
4. Verify that **a** is coprime with 256.
5. XOR each plaintext byte with the previous ciphertext byte (or IV for the first byte).
6. Encrypt the result using the Affine Cipher:
$$C=(a \times X+b) \bmod 256$$
7. Store the ciphertext.
8. Repeat for all bytes.

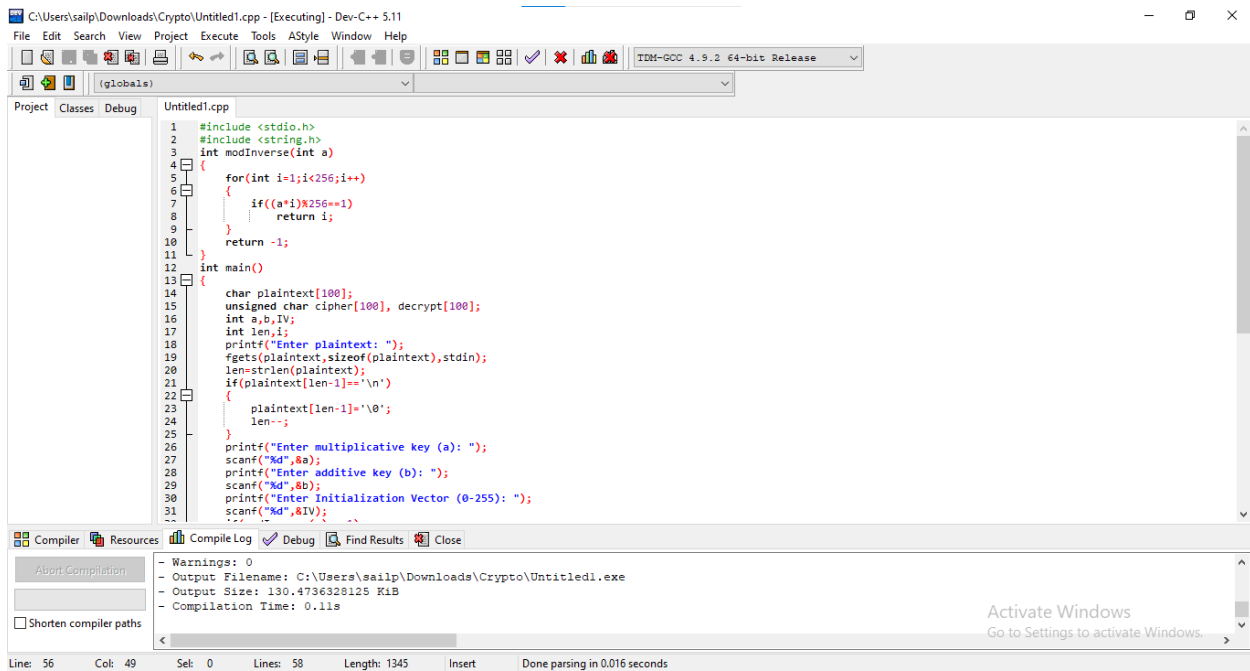
Decryption

9. Compute the modular inverse of **a**.
10. Decrypt each ciphertext byte:
$$X=a^{-1}(C-b) \bmod 256$$
11. XOR the result with the previous ciphertext (or IV) to recover the plaintext.
12. Display the original plaintext.

Result:

Thus, the C program to implement **Cipher Block Chaining (CBC) mode using the Affine Cipher modulo 256** was successfully executed. The plaintext was encrypted into ciphertext and correctly decrypted back to the original plaintext.

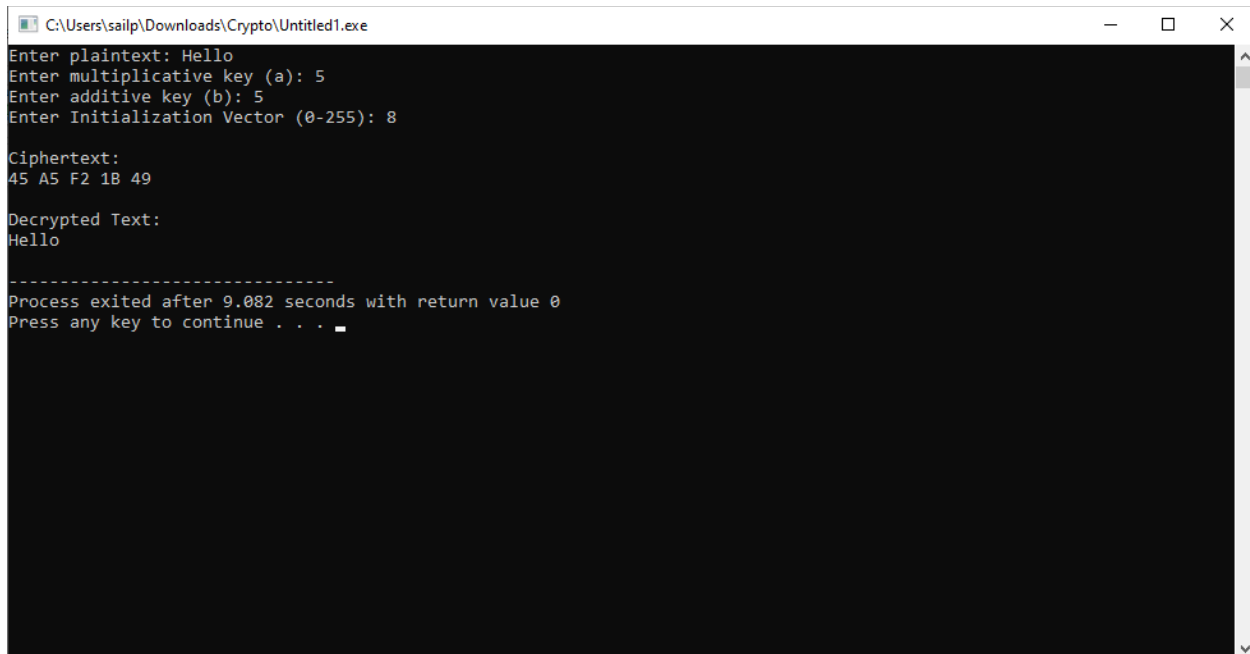
Code:



```
1 #include <stdio.h>
2 #include <string.h>
3 int modInverse(int a)
4 {
5     for(int i=1;i<256;i++)
6     {
7         if((a*i)%256==1)
8             return i;
9     }
10    return -1;
11 }
12
13 int main()
14 {
15     char plaintext[100];
16     unsigned char cipher[100], decrypt[100];
17     int a,b,IV;
18     int len,i;
19     printf("Enter plaintext: ");
20     fgets(plaintext,sizeof(plaintext),stdin);
21     len=strlen(plaintext);
22     if(plaintext[len-1]!='\n')
23     {
24         plaintext[len-1]='\0';
25         len--;
26     }
27     printf("Enter multiplicative key (a): ");
28     scanf("%d",&a);
29     printf("Enter additive key (b): ");
30     scanf("%d",&b);
31     printf("Enter Initialization Vector (0-255): ");
32     scanf("%d",&IV);
33 }
```

Compiler Warnings: 0
Output Filename: C:\Users\sailp\Downloads\Crypto\Untitled1.exe
Output Size: 130.4736320125 KiB
Compilation Time: 0.11s

Output:



```
C:\Users\sailp\Downloads\Crypto\Untitled1.exe
Enter plaintext: Hello
Enter multiplicative key (a): 5
Enter additive key (b): 5
Enter Initialization Vector (0-255): 8

Ciphertext:
45 A5 F2 1B 49

Decrypted Text:
Hello

-----
Process exited after 9.082 seconds with return value 0
Press any key to continue . . .
```